

Continuous Integration and Installation of Tomcat

Step 1:

Open Terminal

- `sudo -i` → Switches the user to root (administrator) mode with full system privileges.
- `sudo apt update && sudo apt upgrade -y` → Updates the package list and upgrades all installed packages automatically without asking for confirmation.

```

sudo: upgrade: command not found
root@vaishnavi-virtualbox:~# sudo apt update && sudo apt upgrade -y
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:4 https://ppa.launchpadcontent.net/mozillateam/ppa/ubuntu jammy InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
E: dpkg was interrupted, you must manually run 'sudo dpkg --configure -a' to correct the problem.

```

Step 2:

Download Apache Tomcat

- `cd /tmp` → Changes the current directory to the system's temporary folder.
- `wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat9.0.95.tar.gz` → Downloads Apache Tomcat 9.0.95 compressed file from the official Apache archive to the system.

```

root@vaishnavi-virtualbox:~# cd /tmp
root@vaishnavi-virtualbox:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-13 19:55:50-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.95.tar.gz'

apache-tomcat-9.0.95.tar.gz      100%[=====>]  12.13M  454KB/s   in 20s
2026-04-13 19:56:11 (634 KB/s) - 'apache-tomcat-9.0.95.tar.gz' saved [12715996/12715996]

```

Step 3:

Create Directory for Tomcat Installation

- `cd ..` → Moves one level up from the current directory.
- `cd /opt` → Navigates to the /opt directory used for installing optional software.
- `mkdir tomcat` → Creates a new directory named tomcat to store the Tomcat files.
- `ls` → Lists directory contents to verify that the tomcat folder is created successfully.

```

root@vaishnavi-virtualbox:/tmp# cd ..
root@vaishnavi-virtualbox:/# cd opt
root@vaishnavi-virtualbox:/opt# ls
google tomcat
root@vaishnavi-virtualbox:/opt# cd tomcat
root@vaishnavi-virtualbox:/opt/tomcat# cd ..
root@vaishnavi-virtualbox:/opt# cd tmp
-bash: cd: tmp: No such file or directory
root@vaishnavi-virtualbox:/opt#

```

Step 4: Extract Apache Tomcat Files

- `sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1` →

Extracts the downloaded Tomcat archive into /opt/tomcat directory by removing the toplevel folder and placing all files directly inside it.

```

apache-tomcat-9.0.95/webapps/host-manager/css/manager.css
apache-tomcat-9.0.95/webapps/host-manager/images/asf-logo.svg
apache-tomcat-9.0.95/webapps/host-manager/images/tomcat.svg
apache-tomcat-9.0.95/webapps/host-manager/index.jsp
apache-tomcat-9.0.95/webapps/manager/META-INF/context.xml
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/401.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/403.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/404.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorCerts.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/sessionsList.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/web.xml
apache-tomcat-9.0.95/webapps/manager/css/manager.css
apache-tomcat-9.0.95/webapps/manager/images/asf-logo.svg
apache-tomcat-9.0.95/webapps/manager/images/tomcat.svg
apache-tomcat-9.0.95/webapps/manager/index.jsp
apache-tomcat-9.0.95/webapps/manager/status.xsd
apache-tomcat-9.0.95/webapps/manager/xform.xsl
apache-tomcat-9.0.95/bin/catalina.sh
apache-tomcat-9.0.95/bin/ciphers.sh
apache-tomcat-9.0.95/bin/configtest.sh
apache-tomcat-9.0.95/bin/daemon.sh
apache-tomcat-9.0.95/bin/digest.sh
apache-tomcat-9.0.95/bin/makebase.sh
apache-tomcat-9.0.95/bin/setclasspath.sh
apache-tomcat-9.0.95/bin/shutdown.sh
apache-tomcat-9.0.95/bin/startup.sh
apache-tomcat-9.0.95/bin/tool-wrapper.sh
apache-tomcat-9.0.95/bin/version.sh

```

Step 5:

Configure Tomcat Server

- `cd /opt/tomcat/conf` → Navigates to the Tomcat configuration directory.
- `ls` → Lists configuration files inside the directory.
- `gedit server.xml` → Opens the main Tomcat configuration file for editing.
- Port Change (8080 → 9090) → Modifies the default Tomcat port to 9090 to avoid conflicts with other services.

```

root@vaishnavi-virtualbox:/opt/tomcat# ls
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
root@vaishnavi-virtualbox:/opt/tomcat# cd conf/
root@vaishnavi-virtualbox:/opt/tomcat/conf# ls
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml

```

```

52 so you may not define subcomponents such as "Valves" at this level,
53 Documentation at /docs/config/service.html
54 -->
55 <Service name="Catalina">
56
57 <!--The connectors can use a shared executor, you can define one or more named thread
58 pools-->
59 <!--
60 <Executor name="tomcatThreadPool" namePrefix="catalina-exec-"
61     maxThreads="150" minSpareThreads="4"/>
62 -->
63
64 <!-- A "Connector" represents an endpoint by which requests are received
65 and responses are returned. Documentation at :
66 Java HTTP Connector: /docs/config/http.html
67 Java AJP Connector: /docs/config/ajp.html
68 APR (HTTP/AJP) Connector: /docs/apr.html
69 Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
70 -->
71 <Connector port="8080" protocol="HTTP/1.1"
72     connectionTimeout="20000"
73     redirectPort="8443"
74     maxParameterCount="1000"
75 />
76 <!-- A "Connector" using the shared thread pool-->
77 <!--
78 <Connector executor="tomcatThreadPool"
79     port="8080" protocol="HTTP/1.1"
80     connectionTimeout="20000"
81     redirectPort="8443"
82     maxParameterCount="1000"
83 />
84 -->
85 <!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443
86 This connector uses the NIO implementation. The default
87 SSLImplementation will depend on the presence of the APR/native

```

Step 6:

Configure Tomcat User Access

- gedit tomcat-users.xml → Opens the Tomcat user configuration file.
- Add user with role → Creates a user (admin) with password and assigns manager-gui role to access the Tomcat Manager dashboard

```

See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
    version="1.0">
    <user username="admin" password="Admin" roles="manager-gui"/>
<!--

By default, no user is included in the "manager-gui" role required
to operate the "/manager/html" web application. If you wish to use this app,
you must define such a user - the username and password are arbitrary.

```

Step 7:

Start Tomcat Server

- cd /opt/tomcat/bin → Navigates to the Tomcat binary directory.
- ls → Lists all available startup and shutdown scripts.
- sh startup.sh → Starts the Apache Tomcat server successfully.

```

root@vaishnavi-virtualbox:/opt/tomcat/cont# cd ..
root@vaishnavi-virtualbox:/opt/tomcat# ls
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
root@vaishnavi-virtualbox:/opt/tomcat# cd bin
root@vaishnavi-virtualbox:/opt/tomcat/bin# ls
bootstrap.jar          ciphers.bat          configtest.bat        digest.sh             setclasspath.sh       startup.sh             tool-wrapper.sh
catalina.bat           ciphers.sh           configtest.sh         makebase.bat         shutdown.bat           tomcat-juli.jar       version.bat
catalina.sh            commons-daemon.jar   daemon.sh             makebase.sh          shutdown.sh            tomcat-native.tar.gz  version.sh
catalina-tasks.xml     commons-daemon-native.tar.gz digest.bat            setclasspath.bat     startup.bat            tool-wrapper.bat
root@vaishnavi-virtualbox:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@vaishnavi-virtualbox:/opt/tomcat/bin# cd ..
root@vaishnavi-virtualbox:/opt/tomcat#

```

Step 8:

Verify Tomcat Installation

- Open Browser (<http://localhost:9090>) → Confirms Tomcat server is running successfully by displaying the default Tomcat page.
- Access Manager (<http://localhost:9090/manager/html>) → Logs into the Tomcat Manager dashboard to manage and deploy web applications.

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.95

If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Server Status](#)

[Manager App](#)

[Host Manager](#)

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Recipes & ASA](#)
- [JDBC Data Sources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the manager webapp is restricted. Users are defined in:

`$CATALINA_HOME/conf/tomcat-users.xml`

In Tomcat 9.0 access to the manager application is split between different users. [Read more...](#)

- [Release Notes](#)
- [Changelog](#)
- [Migration Guide](#)
- [Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/README.txt`

Developers may be interested in:

- [Tomcat 9.0 Bin Directory](#)
- [Tomcat 9.0 JavaDocs](#)
- [Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Sub-headers)
- [tomcat-users](#)
User support and discussion
- [tomcat-dev](#)
User support and discussion for Apache Tomcat
- [tomcat-bugs](#)
Development mailing list, including commit messages

Other Downloads: [Tomcat Downloads](#) [Tomcat Components](#) [Tomcat History](#)

Other Documentation: [Servlet Specification](#) [JSP 2.3 Specification](#)

Get Involved: [Overview](#) [Source Repositories](#)

Miscellaneous: [License](#)

Apache Software Foundation: [About Us](#) [Contact Us](#)

Sign in

http://localhost:9090

Username

admin

Password

Cancel

Sign in



Tomcat Web Application Manager

Message: OK

Manager

List Applications

HTML Manager Help

Manager Help

Server Status

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	Yes	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/static	None specified	Tomcat Static Resources	Yes	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/examples	None specified	Servlet and JSP Examples	Yes	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	Yes	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/manager	None specified	Tomcat Manager Application	Yes	1	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes

Deploy

Deploy directory is /usr/local/tomcat/webapps

Context Path

Version (for parallel deployment)

XML Configuration file path

URL to Deployable Code